

Question (from image_efdc60.png):

- Explain cache read hit/miss. / Define miss penalty, cache miss, hit rate, tag in caching. / Briefly explain the characteristics and working mechanisms of Cache Memory in terms of:
a) Speed and Location, b) The Principle of Locality, c) Hits and Misses

Answer:

1. Core Characteristics & Working Mechanisms

- **Speed and Location:** Cache is a tiny, ultra-fast memory located inside or right next to the CPU chip, designed to bridge the speed gap between the fast processor and slower main memory (RAM).
- **The Principle of Locality:** Caching relies on the fact that programs tend to reuse recent data.
 - *Temporal Locality:* If data is used once, it will likely be used again soon.
 - *Spatial Locality:* If data is used, nearby data in memory will likely be used next.

2. Key Cache Terminology Definitions

- **Cache Hit (or Read Hit):** Occurs when the CPU looks for data and finds it successfully in the cache, resulting in an instant data delivery.
- **Cache Miss (or Read Miss):** Occurs when the requested data is not found in the cache, forcing the system to search the slower main memory.
- **Hit Rate:** The percentage of all memory accesses that are successfully found in the cache.
- **Miss Penalty:** The extra time (in clock cycles) required to fetch a block of data from main memory and replace it in the cache when a miss occurs.
- **Tag:** A unique identification field attached to each cache row that indicates the actual address of the main memory block currently stored in that slot.

Question (from image_efdfc7.png):

- Describe the following with example: a) Multi-level Cache / Q6: Briefly explain the Cache Memory Hierarchy levels: L1 Cache, L2 Cache, and L3 Cache.

Answer:

- **Multi-level Cache:** It is a memory architecture that uses a hierarchy of multiple cache layers (L1, L2, L3) with different sizes and speeds to balance cost, performance, and processor waiting times.

Cache Hierarchy Levels

- **L1 Cache (Level 1):**
 - *Characteristics:* The smallest, fastest, and closest cache to the CPU core. It usually operates at the exact speed of the processor.
 - *Example:* A typical CPU core might have a **32 KB** L1 Data cache built directly inside it.
- **L2 Cache (Level 2):**
 - *Characteristics:* Slightly larger and slower than L1. It acts as a secondary buffer to catch any data requests that miss the L1 cache.
 - *Example:* A dedicated **512 KB to 1 MB** L2 cache assigned per individual CPU core.
- **L3 Cache (Level 3):**
 - *Characteristics:* The largest and slowest cache layer among the three, but still significantly faster than system RAM. It is shared across all processor cores.
 - *Example:* A shared **16 MB to 32 MB** pool of L3 cache accessible by all cores on the chip.

Question:

- Q3: What are the core steps to handle a cache miss when a processor requests data?

Answer:

When a cache miss occurs, the CPU must fetch the data from the slower main memory (RAM) through the following core steps:

1. **Send Address:** The CPU sends the missed memory address directly to the Main Memory Controller.
2. **Fetch Data:** Main memory reads the entire data block containing the requested word.
3. **Write to Cache:** The fetched data block is written into the appropriate slot inside the cache hierarchy, and its unique Tag and Valid bit are updated.
4. **Deliver to CPU:** The specific word requested by the processor is delivered to the CPU registers, and the instruction resumes execution.

Question (from image_efe400.png):

- Describe the following with example: b) 3Cs model.

Answer:

The 3Cs model categorizes the reasons why a cache miss occurs:

- **1. Compulsory Misses (Cold-Start):**
 - Happens the very first time a block of data is accessed by the processor, as it cannot possibly be in the cache yet.
 - *Example:* Opening a brand new application right after booting up your computer.
- **2. Capacity Misses:**
 - Happens when the cache is simply too small to hold all the active data blocks needed by the running program.
 - *Example:* Trying to run a heavy data loop that requires 128 KB of memory inside a tiny 64 KB cache.
- **3. Conflict Misses (Collision):**
 - Happens in direct-mapped or set-associative caches when multiple active memory blocks compete for and map to the exact same cache slot, kicking each other out.
 - *Example:* A program repeatedly bouncing between two data variables that both share index location 5 in the cache.

Question (from image_efe747.png):

- Caches

Answer:

- **What it is:** A cache is a small, ultra-fast hardware memory layer located close to or inside the CPU core.
- **Purpose:** It temporarily stores frequently accessed data and program instructions from the slower main memory (RAM).
- **How it helps:** It bridges the massive speed gap between the fast processor and slower RAM, significantly speeding up overall system execution.